Requirements Engineering Challenges in Developing Real-World Information system

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Abstract - Requirement Engineering is a very important phase in software development. It was focused on understanding the needs of users and clients and then turning it into system design.However,in actual software development, if requirement engineering is not done well, it may lead to project failure and at the same time increase the cost. In this article, three common problems in RE, difficulty in communication, frequent change in requirement and unclear requirement expression had been analysed via a systematic literature review. Moreover, when requirements are described in natural language can also cause communication misunderstandings. Besides, research has found that traditional methods gathering requirements might overlook some important non-functional requirements such as security. At the same time, if changes in requirements are not properly managed, they can also affect the software reliability. Hence, this articles suggests a few ways to improve and reduce the issues in requirement engineering and software development quality.

Keywords—database design, software quality, performance efficiency, ISO/IEC 25010

I. INTRODUCTION

Requirement Engineering (RE) is known as a initial process where it brings an essential part to start a project without a formal ideas.A document of requirement specification system that contains information such functional requirement and non-functional requirement are falls under information that must be included in this phase (*Pohl, K., 1996*).It ensures that the developed system fulfill the requirement and expectations of its stakeholders, including users, administrator, and developers (SWEBOKwiki, 2015).In short,a successful system development stem from a excellent Requirement Engineering (Järvinen, R. (2026).However,common challenges still exist throughout this process.

II. PROBLEM STATEMENT

According to the study,the problem faced by RE happened during communication phase between RE and stakeholder.Stakeholder may be the one who decided to make engagement between both parties. In reality, poor communication between stakeholders and RE can lead to an inaccurate interpretations and misunderstanding of system requirements (*Mello, O. C. and Fontoura, L. M, 2022*). Another problem is incomplete requirement. One factor that lead to this case is uncertain requirements during the specification phase. In usual case, to identify the requirement before develop can be categorized as difficult thing due to several factors. Stakeholders may have different expectations,limited knowledge in the scope or difficult to express their actual needs (*Mello, O. C. and Fontoura, L. M, 2022*).

III. RESEARCH OBJECTIVES

Research Objectives

1.To examine how communication between stakeholder and RE affect the development of information systems.
2.To explore practises that can help prevent incomplete requirement from stakeholder.

Research Questions

1.How do communication problems between stakeholder and RE affect the development of information systems.
2.What practice can be adopted to prevent incomplete requirement from stakeholder.

IV. LITERATURE REVIEW

First and foremost, Requirements Engineering (RE) functions as the important bridge between business objectives and technical software implementation (Nuseibeh and Easterbrook, 2000; Sommerville, 2016). However, studies

report that insufficient requirement engineering practices are still a main reason which indicate why the project will fail and have structural defects as well as over budget in the real-world environments (Méndez Fernández et al., 2017)..

Requirement Gathering and Communication Barriers

The primary obstacle during the requirement gathering process comes from the communication gap between the non-technical experts and software developers (Nuseibeh and Easterbrook, 2000; Inayat et al., 2015) . Business stakeholders often have professional knowledge of operational insights and experience that they take for granted and fail to explain clearly during the initial interviews (Sommerville, 2016). Misunderstandings always happen during communication. It is because of the use of domain-specific terms and conflicting needs between different stakeholder groups (Inayat et al., 2015; Cao and Ramesh, 2008) . A research study shows that when software teams rely only on traditional interview techniques instead of visual prototypes or domain modeling, key non-functional requirements such as security and scalability will tend to be systematically ignored (Méndez Fernández et al., 2017; Umar and Lano, 2024).

Changing of Requirements and Scope Creep

Nowadays, in modern software development, requirements usually do not stay the same. As the market environment and regulations as well as the user preference change, software requirements might keep being added, modified and also removed (Farwah and Rai, 2021). However, if these changes do not manage properly, it can make the scope creep happen easily. Hence, projects will be delayed and also increase the complexity or maintenance difficulty of the system architecture (Farwah and Rai, 2021). Besides, even though agile development allows developers to accept the changes flexibly during the development process, the research shows that if they do not have a clear management or frequent modification will lead to reduction of software reliability and increase of cost maintenance (Inayat et al., 2015; Cao and Ramesh, 2008). In conclusion, it is normal to modify the software requirement frequently but if it was not managed properly then it will bring a big negative effect to the software.

The ambiguity of natural language specifications

In the industry, most of the system requirements are written in natural language which is plain text because it allows clients and non-technical people to understand easily (Sommerville, 2016) . However, natural language is not always accurate. The same sentences can be understood in multiple ways which can easily lead to misunderstanding between the developer and testers (Umar and Lano, 2024; Inayat et al., 2015) . Moreover, if this misunderstanding occurs during the requirement analysis phase, these problems can persist into the development and testing phase later. Not only that, research shows that fixing a requirements error during the system maintenance phase can cost up to 100 times more than during the requirement gathering phase (Umar and Lano, 2024; Umar and Lano, 2024) . Therefore, recent studies suggest combining structured natural language templates with semi-formal UML models to make requirement expression clearer and at the same time reduce the differences in understanding different people (Sommerville, 2016; Umar and Lano, 2024).

V. RESEARCH METHODOLOGY

This research uses a Systematic Literature Review approach to systematically study the challenges that information systems in real life face in terms of requirement engineering. It referred to standard research guidelines in the field of software engineering and gathered relevant materials from several online resources including IEEE, Xplore, and ACM Digital Library. Elsevier, SpringerLink and Wiley.

Besides, this research set clear inclusion and exclusion criteria to select high quality of reliable research resources. The resources included had to be published between 2000 and 2026 and also be either peer-reviewed journal articles or IEEE conference papers. Also, the content has to be related with the challenge faced in the requirement engineering as well as include the survey. Instead, the other articles that do not meet the requirement will be excluded.

Moreover, this study organizes and analysis the collected resource by categorized the main

challenge into one group. Then, through the resource, researchers find common issues that happen during the software development phase such as difficulty in collecting requirements, frequent change in requirement as well as lack of management. Then, the researchers compared these and summarised it to understand the cause of each problem and how its effect was. Finally, a solution will be proposed to help in reducing the gap between the clients and the system.

VI. DISCUSSION

Documentation

Documentation often involves Analysis, Processing and Comparison and one of the main reasons that the project could go off track by lacking documentation. Having documentation that is not accessible to all parties involved, that is not managed properly are a major risk factors for a project. It is necessary to have good practices defined to document and manage the documents to ensure the software product quality.

Without requirement Engineering

Humans can't manage to process thousands of pages of documents to search for relevant information every time they are required to do the project since it is not realistic and a waste of time . Delays often happen because of misunderstanding and misinterpretation during the search for the relevant information. There are thousands of documents and not every document has relevant information for the project, therefore misunderstanding and misinterpretation may occur. Some documents also have errors, and if the teams overlook the error, it can result in costly corrections later in the development process.

With requirement engineering

Using an AI-powered system, AI can extract requirements from large documents, such as thousands of documents, to provide relevant information for the team members that are involved in the project. Errors can be avoided during the project since it can automatically detect differences between documents or versions and therefore the team members can search for accurate information (*Reichel ,M.)* . Another option is creating a central hub that is easily accessible to all team members and stakeholders, allowing them to view software requirements and other development information. *(Taller, H.)* .

VII. CONCLUSION

Communication problems are widely recognized and it can occur between stakeholder and requirement engineering. It can cause a major factor in the delay and failure of software projects. Furthermore, the specification documentation may be so large that no member of the software team has read it at all which can cause misunderstanding. Informal communications need to be encouraged between all stakeholders. Face-to-face discussion can prevent the communication problem from happening. To prevent incomplete requirements from stakeholders, it is better to ask them for stories rather than hearing their requirements such as "We need a better … [tool].". User stories work because they mirror how humans think. We're building the system to provide convenience for people to do their jobs rather than for requirements documents only.

VIII. REFERENCE

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IX. APPENDICES

GitHub Logbook: https://github.com/users/nasaaie53/projects/5/views/1